Ahmed M. A. Sayed

a.k.a. Ahmed M. Abdelmoniem

Lecturer (Assistant Professor)
Director of BDS Programme

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Research Profile

I lead SAYED Systems Group (`https://sayed-sys-lab.github.io`) in which we build Scalable, Advanced Yet Efficient Distributed Systems of the Future. Our research spans inter-related disciplines of computer science and engineering with a focus on system design and optimization for machine learning systems (training and inference efficiency, distributed ML, federated learning), distributed systems (architecture design, performance analysis, resource allocation, algorithmic optimization), computer networks (traffic engineering, congestion control, performance optimization, software-defined networking), and wireless networks (routing in mobile ad-hoc and wireless sensor networks).

Current Research Interests

- Systems for ML & ML for Systems
- Federated & Distributed Machine Learning
- Computer Networks and Systems

Grants and Funding

2023 **PI - EPSRC - UKRI**, *KUber: Knowledge Delivery System for Machine Learning at Scale*, Granted 650K GBP

2022 **CoI - EPSRC - REPHRAIN Center**, *Moderation in Decentralised Social Networks (DSNmod)*, Granted 81K GBP

2022 **CoI - HKRGC - GRF**, *ML Congestion Control in SDN-based Data Center Networks*, Granted 600K HKD

2021 **CoI - KAUST - Competitive Research Grant**, *Machine Learning Architecture for Task-based Information Transfer*, Granted 400K USD

2013 **PhD - HKPFS - HK Research Grand Council**, *Efficient Optimizations for Water Pipes Monitoring Wireless Sensor Networks*, Granted - 150K USD

Education

2013-2017 **Ph.D., Computer Science and Engineering (GGA 4.0/4.2)**, *Computer Science and Engineering Department, Hong Kong University of Science and Technology*, Hong Kong

PhD Thesis

Title *On Improving The Performance of TCP Applications in Public Cloud Networks*

Supervisor Assoc. Prof. Brahim Bensaou (CSE, HKUST)

Committee Prof. Gary Chan (CSE, HKUST), Prof. Danny Tsang (ECE, HKUST), Assoc. Prof. Kai Chen (CSE, HKUST), Assoc. Prof. Chun Tung Chou (UNSW, Australia)

Description Proposing efficient schemes that have demonstrated to achieve considerable performance gains for cloud applications via mathematical modeling, empirical analysis, simulation, hardware prototyping and real-testbed implementation and experiments on various network scenarios and topologies.

2008–2012 **M.Sc., Computer Science (Distinction, ranked 1^{st})**, *Computer Science Department, Faculty of Computers and Information, Assiut University*, Egypt

Masters Thesis

Title *Routing Optimization of Mobile AD-HOC Networks Based on Ant Colony Algorithms*

supervisor Prof. Hosni Ibrahim, Prof. Marghany H. Mohamed, Asst. Prof. Abdel-Rahman Hedar

Description Proposing efficient routing optimizations based on Ant Colony Algorithms (ACO) that have demonstrated to achieve considerable performance gains for Mobile Ad-Hoc Networks (MANET) routing protocols via mathematical modeling, simulation analysis and implementation on various network scenarios and topologies.

2003–2007 **B.Sc., Computer Science (Distinction with Honors, ranked 2^{nd})**, *Computer Science Department, Faculty of Computers and Information, Assiut University*, Egypt

Bachelors Thesis

Title *Enabling Video Calls over Internet and Bluetooth* - Ministry of Telecommunications, Egypt

Supervisor Prof. Yousef B. Mahdi

Description Full system implementation and delivery of a fully functional system for enabling video calls over bluetooth and Internet using C# as back-end server and J2ME as the front and back-end on various camera-ready mobile devices.

2000–2003 **Egyptian General Secondary Examination (98.2%, ranked in the top 200 out of 400K+ student)**, *Almoshir Ahmed Esmail Secondary School*, Assiut, Egypt

Work Experience

2021–Now **Lecturer (Assistant Professor) & Director, MSc Big Data Science Program**, *School of Electronic Engineering & Computer Science, Queen Mary University of London*, UK

Duties **“Research:”** Managing research teams and active grants. Conducting scientific research in computer science and publishing high quality papers, supervising M.S./Ph.D. students and preparing research grant proposals. **“Teaching:”** Teaching and organizing and delivering modules at both the undergraduate and postgraduate levels. Performance assessment of students in weekly self-practice tasks. Interacting with students within office hours. Preparing and mentoring written examination and course projects. **“Administration:”** Responsibility for developing and maintaining the BDS Programme activities in relation to areas of interest and innovation within the Faculty. Active membership in school meetings to discuss departmental and school matters (teaching, research and school management).

Modules Module organizer and sole lecturer for ***ECS765P - Big Data Processing*** module taught as part of the MSc Data Science Program for the postgraduate level (level-7)

Co-organize and teach the ***ECS637U/ECS757P - Digital Media and Social Networks*** module for both undergraduate (level-6) and postgraduate (level-7) levels

2020–2021 **Research Scientist**, *Extreme Computing Research Center (ECRC), King Abdullah University of Science and Technology (KAUST)*, Saudi Arabia

Duties Develop and write research proposals for obtaining research funding grants. Supervise graduate students. Collect resources and work with team members to complete the research plan. Present research results at meetings and engages in result publications. Be up-to-date with new research directions and findings by attending research seminars, workshops, and conferences. Review literature and analyze their results. Conduct system research in a systematic manner with a focus on improving performance, scalability, and interpretability of distributed machine learning. Find efficient solutions for challenging computer science research problems. Design experiments to verify existing and new solutions and techniques. Write papers on novel ideas and new findings and publish them in reputed venues and journals. Communicate ideas and results with members. Understand relevant literature.

2019–2020 **Post-Doctoral Research Fellow**, *Extreme Computing Research Center (ECRC), King Abdullah University of Science and Technology (KAUST)*, Saudi Arabia

Duties Find efficient solutions for challenging computer science research problems. Design experiments to verify existing and new solutions and techniques. Collect resources and work with team members to complete the research plan. Develop and write research proposals for obtaining research funding grants. Present research results at meetings and engage in result publications. Attend research seminars, workshops and conferences to be aware of the new research directions and findings. Review literature and analyze their results. Conduct computer and network system research with focus on improving performance, scalability and interpretability of distributed machine learning applications. Prepare manuscripts of novel ideas and findings to publish them in reputed venues and journals. Communicate ideas and results with members. Understand relevant scientific literature. Work with group members.

2018– **Assistant Professor**, *Faculty of Computers and Information, Assiut University*, Egypt

Duties **"Research:"** Conducting scientific research in computer science and publishing high quality papers, supervising M.S./Ph.D. students and preparing research grant proposals. **"Teaching:"** Teaching and managing Computer Science courses at both the undergraduate level such as ***Software Engineering***, ***Project Management***, ***Software Development and Technical Practice***, ***Computer Security*** and ***Parallel Computing*** and postgraduate level such as ***Object-Oriented Software Engineering*** and ***Intro to Computers - Fine Arts***. Performance assessment of students in weekly self-practice tasks. Interacting with students within office hours. Preparing and mentoring written examination and course projects. **"Administrative highlights:"** Active membership in faculty committee meetings to discuss departmental and school matters (teaching, research and school management). Active membership in department and school councils. Organizing research seminars. Director of Information Technology unit of the school. Membership in research and library council committees.

2017-2018 **Senior Researcher**, *Future Network Theory Lab. Huawei Technologies Investing Co. Ltd.*

Duties Conducting advanced research in network control, traffic engineering and resource management. Leading the system research directions. Research proposals preparation. Research publications. Patent development. Collaboration work management. Recruitment.

Contributions I have contributed with our team in the redefinition and formulation of the ADN vision and its system implementation and evaluation both in simulation and real-testbed environment. ADN (and specifically fast-slow control) is a major long-term project in the lab which is the main focus of the lab and so far resulted in few paper and patent submissions. Have participated and taken lead on several projects, few of the projects are direct collaboration with renowned professors in Stanford, CalTech, Cornell, CUHK, CityU, HKUST, Nanjing, and Tsinghua Universities. Moreover, finished a stalled project in a record time. Other projects involved working closely with 4 interns to bring them to completion. Been actively responsible for the recruitment process for new hires and interns to our lab during different venues (e.g., SIGCOMM & NSDI & INFOCOM, etc). Successfully recruited for the Lab 3 new full-time hires and 1 intern.

2012–2018 **Assistant Lecturer**, *Faculty of Computers and Information, Assiut University*, Egypt

Duties Conducting scientific research in computer science and publishing high quality papers. Pursuing my Doctorate degree. Teaching in classes and laboratories ***Computer Networking***, ***Network Programming*** and ***Network Analysis and Design*** to undergraduate students. Performance assessment of students in weekly self-practice tasks. Interacting with students within office hours. Mentoring and grading written examination and course projects

2018-2021 **Technical Advisor**, *DeepCloudAI - a Decentralized AI-Driven Cloud Computing Infrastructure*

Duties Advising throughout the development of DeepCloudAI (`www.deepcloudai.com`) project, including but not limited to, advising the development of white paper and technical paper, advising the development of the prototype. Consultation on the technical questions and technical problem solutions. Advising on the technical-related activities that is required for the completion of the Initial Coin Offering campaign.

Fall 2017 **Teaching Assistant**, *Department of Computer Science and Engineering, The Hong Kong University of Science and Engineering.*

Duties Assist in teaching ***Computer Networks: An Internet-Perspective - CSIT5610*** MSCIT course to post-graduate students. Grading examination papers. Students' consultation. Examination proctoring.

Spring 2017 **Linux Kernel Network Programming Assignment/Quiz Handler**, *Department of Computer Science and Engineering, The Hong Kong University of Science and Engineering.*

Fall 2016 **Teaching Assistant**, *Department of Computer Science and Engineering, The Hong Kong University of Science and Engineering.*

Duties Assist in teaching ***Computer Networks - COMP 5621*** PG Core and ***Computer Networks: An Internet-Perspective - CSIT5610*** MSCIT course to post-graduate students. Grading examination papers. Students' consultation. Examination proctoring.

2007–2012 **Teaching Assistant**, *Faculty of Computers and Information, Assiut University*, Egypt

Duties Conducting scientific research in computer science and publishing high quality papers. Pursuing my Masters degree. Teaching in classes and laboratories ***Computer Networking***, ***Network Programming***, ***Object-Oriented Programming using C++***, ***Introduction to JAVA Programming***, ***Introduction To Computers***, ***Software Testing***, ***Data Structure***, ***Algorithms***, ***Artificial Intelligence***, ***Operating Systems***, ***Distributed Database*** and ***IT Project Management*** to undergraduate students. Performance assessment of students in weekly lab tasks. Interacting with students within office hours. Mentoring and grading written examination and course projects

Relevant Research Experience and Interests

- Computer Networking, Congestion Control, and Traffic Engineering
- Software Defined Networking and Network Function Virtualization
- Feedback Control of Hybrid and Switched Systems
- Mobile Ad-hoc Networks and Wireless Sensors Network
- AI Optimization and Genetic algorithms

Sample Projects

- ML Congestion Control in SDN-based Data Center Networks, 2022 – Now
- Moderation in Decentralised Social Networks (DSNmod), 2022 – Now
- Machine Learning Architecture for Task-based Information Transfer, 2021 – Now
- Efficient Decentralized Learning in Heterogeneous Mobile Edge Computing, 2020 – Now
- Resource Efficient Federated Learning, 2020 – Now
- Design, Implementation and Analysis of Methods to Improve Performance of Distributed Machine Learning Systems, 2019–2020
- Implementation, Evaluation and Analysis of Fast-Slow Network Control Framework for Application-Driven Networking, 2017–2018
- Implementation, Evaluation and Analysis of Efficient, Scalable and Easily-Deployable Congestion Traffic and Control Schemes in Data Centers and Cloud, 2013–2017

- Design, Implementation and Analysis of various routing protocols optimized for Mobile Ad-Hoc Networks (MANET) via leveraging Artificial Intelligence algorithms inspired from Ant Colonies, 2008–2012
- Implementation of a complete system for enabling video calls over bluetooth and Internet using C# as the back-end server and J2ME as the mobile front-end, 2007

Student Supervision

2024-Now **PhD - QMUL**, *Advancing NLP via Accelerated Deep Learning*

2023-Now **PhD - QMUL**, *Novel Optimizations for Large Audio Models (LAMs)*

2023-Now **PhD - QMUL**, *Efficient Optimizations for Decentralized Cross-Silo Federated Learning*

2022-Now **Intern - QMUL**, *Energy-Aware Methods for Federated Learning on Battery-limited Clients*

2022-Now **Intern - QMUL**, *Novel Optimization for Efficient and Robust Federated Learning*

2021 **MS/PhD - KAUST**, *Mitigating Device Heterogeneity in Federated Learning via Asynchronous Stale Updates*

2021 **MS/PhD - KAUST**, *Prioritizing Participant Selection for Efficient Federated Learning*

2020 **MS/PhD - KAUST**, *Identifying the Limits of Gradient Sparsification Methods for Distributed Machine Learning*

2020 **Research Student Inters - KAUST**, *Study of Fairness and Bias in Federated Learning settings*

2020 **Research Student Inters - KAUST**, *An Efficient compression technique to reduce Communication in Distributed Deep Learning*

2019 **MS/PhD - KAUST**, *Survey and Empirical Analysis of Compressed Communication for Distributed Deep Learning*

2019 **MS/PhD - KAUST**, *Theoretical and Empirical Analysis of Layerwise and Whole-Model Compressed Communication Methods in Distributed Machine Learning*

2019 **Research Student Intern - KAUST**, *Energy-Efficiency of Hardware Offloading: Case-Study on Distributed Machine Learning*

2019 **Research Student Intern - KAUST**, *Scaling Distributed Machine Learning with In-Network Aggregation using Smart NICs*

2019 **Research Student Intern - KAUST**, *Accelerating Distributed Deep Learning with Adaptive Compression and Communication Scheduling*

2018 **PhD Research Student Intern - Huawei Research**, *Leveraging Programmable Data Plane to Accelerate Distributed Applications*

2018 **PhD Research Student Intern - Huawei Research**, *An Online Learning Multi-Path Selection Framework for Multi-path Transmission Protocols*

2018 **Research Student Intern - Huawei Research**, *Implementation of an SDN-based Fast-Slow Control system to Realise an Operational Prototype of the Application-Driven Networking (ADN) Framework*

2007-2013 **FYPs Undergraduate Students - Assiut University**, *Management System for controlling Wireless Access Points, HoneyPot Server Application, WiiMote Body Tracking & Robot Control System, Steganography Application to hide data in images and videos, Remote Desktop Control using Mobile Phones, Mobile Application in Traffic Service, Tourist Heaven - a tourist social networking application and Egyptian tourism company web system*

Honors, Awards and Scholarships

2013–2017 **Hong Kong PhD Fellowship (HKPFS) award**, *HK Research Grants Council*

2013–2017 **HKPFS research travel grant award**

2017	**Student Participation Grant**, *Local Computer Networks (IEEE LCN)*, IEEE CompSoc
2015	**Travel Grant award**, *Global Communications (GlobeCom) conference*, IEEE ComSoc
2007	**FYP sponsorship award**, *Ministry of Telecommunications*, Egypt
2003-2007	**Undergraduate Distinction award**, *for TGA of 85%-above*, Assiut University
2003–2007	**Dean's Honors**, *TGA of 85%+*, Faculty of Computers and Information, Assiut University

Programming, Presentation, Modeling and System Skills

Presentation	Power Point, Beamer — Typography: LaTeX, MicroSoft office
AI/ML	Tensorflow, PyTorch, Distributed ML, Federated Learning, RAY, Horovod, BytePS, WANDB
Modeling	Mathematical, Queueing theory, MatLab, Simulink
Info. Extraction	Awk, Gnuplot, Matplotlib, Pandas, Bokeh, Perl, Excel, SPSS
Simulation	NS-2, NS-3, OPNET Modeler, Mininet
Programming, Web & DB	C, C++, Java, Python, C#, Git, Network Programming, SDN Applications, Ryu SDN Controller, P4/Tofino, Verilog, Tcl, shell scripting, HTML, PHP, SQL Server, Oracle DB
System Prototyping	Linux Kernel Modules, NetFilters, TCP/IP Stack, RDMA/RoCE/PFC, NetFPGA, Kubernetes, Kprobes, Open vSwitch, OVN, OpenFlow, OpenStack, DPDK, JUJU, CONDA, Ubuntu MASS, Broadcom OpenNSL & OFDPA API, Amazon EC2, Microsoft Azure
Virtualization	KVM, Qemu, Xen Hypervisor, Virtual Box
OS	Linux (Desktop/Server), Windows (Pro/Server), MacOS, Open Network Linux, SONiC

Language Proficiencies

Arabic	Native	*Listening, Reading, Speaking and Writing*
English	Fluent	*Listening, Reading, Speaking and Writing*

Professional Membership

IEEE	IEEE and IEEE ComSoc Society Regular Member
ACM	ACM Regular Member
USENIX	USENIX Regular Member
Cisco	Cisco Networking Academy Certified Instructor
Comp Soc	Hong Kong Computer Society Member

Professional Courses and Certification

Nvidia	NVIDIA training: Data Science at Scale
Nvidia	Nvidia-HKUST (DBI) Joint Deep Learning Workshop
Cisco	Cisco Certified Networking Associate (CCNA)
Oracle	Oracle Database Administration (Oracle DBA)

Surplus Courses

QMUL	PGCAP - ADP7216 - Learning and Teaching in Higher Education	
	PGCAP - ADP7217 - Learning and Teaching in the Disciplines	
	PGCAP - ADP7218 - Curriculum Design	
	PGCAP - ADP7219 - Action (Practitioner) Research	
KAUST	Industrial Course on Deep Learning	
HKUST	ELEC4320 - FPGA-based Design	ELEC4010G - Control System Design

	COMP4511 - System and Kernel Programming in Linux
	COMP6611A - Data Center Networks and Cloud Computing
	COMP6611B - Cloud Computing and Data Analysis Systems
	FINA6900N - Startup Financing and Operations
	IELM5570 - Network Optimization in Transport Systems
Stanford	CVX101 - Convex Optimization
MIT	6.00.1x - Introduction to Computer Science and Programming Using Python
UPValencia	DC201x - Dynamics and Control
Microsoft	DAT202.1x - Processing Big Data with Hadoop in Azure HDInsight

Professional Development Courses

QMUL	Challenging Unconscious Bias	Equality and Diversity Briefing
	Anti Bribery Essentials	Safeguarding Essentials
	GDPR for Staff	Cyber Security Training
KAUST	Scientific Writing	Communicating with Confidence
	Effective Scientific Writing	Research Communication Skills
	The Art and Science of Communication	Proven Techniques for Technical Communication
	Conveying messages with graphs	Making the most of your presentation
	The art and science of communication	Proven techniques for technical communication
HKUST	Effective Teaching Skills	Marking & Grading
	High-Tech Entrepreneurship	Understand the World of Work
	What Takes to be a Good Researcher	Conducting Labs
	Effective Presentation for Teaching	How to Write a Journal Paper
	How to Write CS Papers	How to Get Published
	Research ethics: Communities, choices, and values	
	Balancing Time between TA Duties and Research	
	Presenting Myself Through a Winning Profile	
AUN	Effective and Efficient Presentation	Academic Work and Research Ethics
	Scientific Publishing	Teaching Methodologies and Skills
	Credit Hour System	Time and Meeting Management
	Quality Assurance of Education Process	
	Communication Skills for Different Educational Approaches	

Voluntary Services

Editor	Frontiers in HPC on the research topic of HPC for AI in Big Model Era
Program Chair	5th International Workshop on Embedded and Mobile Deep Learning (EMDL) co-located with ACM MobiSys, Virtual Online, 2021
	KAUST-NeurIPS Workshop on Advances of Machine Learning, KAUST, 2019
Tutorial	Distributed Deep Learning Clinic, KAUST, Saudi Arabia, 2020
	Data Analytics in the Cloud in the International BioDialog Project, Exhibition and Hackathon on BioDiversity Informatics, Egypt, 2018
Preview	Congestion Control Session, ACM SIGCOMM, 2022
Reviewer	ACM/IEEE Transactions on Networking (ACM/IEEE ToN)

IEEE Transactions on Neural Networks and Learning Representation (IEEE TLNNLS)
IEEE Internet of Things Journal (IEEE IoTJ)
IEEE Journal on Selected Areas in Communications (IEEE JSAC)
ACM Transactions on Modeling and Performance Evaluation of Computing Systems
IEEE Transactions on Cloud Computing (IEEE TCC)
IEEE Transactions on Mobile Computing (IEEE TMC)
IEEE Transactions on Network and Service Management (IEEE TNMS)
IEEE Access
Journal of Computer Networks, Elsevier
Journal of Computer Communications, Elsevier
Journal of Future Generation Computer Systems, Elsevier
Journal of King Saud University - Computer and Information Sciences, Elsevier
Journal of Computer Standards and Interfaces, Elsevier
Journal of Telecommunication Systems, Springer
International Journal of Artificial Intelligence (IJ-AI)
The Computer Journal by Oxford Academic
The Arabian Journal for Science and Engineering (AJSE)
International Journal of Systems, Control and Communications (IJSCC)

TPC/Reviewer ACM Conference on emerging Networking EXperiments and Technologies (CoNext)
IEEE International Conference on Distributed Computing Systems (ICDCS)
European Workshop on Machine Learning and Systems (EuroMLSys), ACM EuroSys
International Conference on Machine Learning (ICML)
ACM Workshop on Data Privacy and Federated Learning Technologies for Mobile Edge Networks (FedEdge), ACM MobiCom
IEEE Conference on High-Performance Switching and Routing (IEEE HPSR)
IEEE Vehicular Technology Conference (IEEE VTC)
USENIX Annual Technical Conference (ATC)
AAAI Conference On Artificial Intelligence (AAAI)
IEEE Conference on Network Protocols (IEEE ICNP)
IEEE International Performance Computing and Communications Conference (IPCCC)
IEEE Conference on Local Computer Networks (LCN)
ACM Conference on Modeling and Simulation of Wireless and Mobile System (MSWiM)
Workshop on Machine Learning for Software Networks (NetLearn), IEEE NetSoft

References - available upon request

Assoc. Prof.	Brahim Bensaou	*CSE Dept., HKUST, brahim@cse.ust.hk*
Assoc. Prof.	Marco Canini	*CEMSE Division, KAUST, macro@kaust.edu.sa*
Assoc. Prof.	Kai Chen	*CSE Dept., HKUST, kaichen@cse.ust.hk*
Dr.	Bai Bo	*Huawei's Theory Lab, ee.bobbai@gmail.com*

Research Accomplishments-see Google Scholar

Thesis

- **Ahmed M. Abdelmoniem**, **"On Improving the Performance of TCP Applications in Public Cloud Networks"**. Ph.D. Thesis, HKUST, Hong Kong, `https://lbezone.ust.hk/bib/991012554564103412`, 2017.

- **Ahmed M. Abdelmoniem**, **“Routing Optimization of Mobile AD-HOC Networks Based on Ant Colony Algorithms”**. M.Sc. Thesis, Assuit University, Egypt, `http://www.aun.edu.eg/thesis_files/4341.pdf`, 2012.

Published International Refereed Journal Publication

- Z. Tian, Z. Wang, **Ahmed M. Abdelmoniem**, G. Liu and C. Wang, **“Knowledge Representation of Training Data With Adversarial Examples Supporting Decision Boundary”**, in *IEEE Transactions on Information Forensics and Security, vol. 18, pp. 4116-4127, 2023.*
- **Ahmed M. Abdelmoniem** and Brahim Bensaou. **“Enhancing TCP via Hysteresis Switching: Theoretical Analysis and Empirical Evaluation”**. *ACM/IEEE Transactions on Networking (ToN), 2023.*
- **Ahmed M. Abdelmoniem** and Chen-yu Ho, Pantelis Papageorgiou, Marco Canini. **“A Comprehensive Empirical Study of Heterogeneity in Federated Learning”**. *IEEE Internet of Things (IoT) Journal, 2023.*
- Amuda James Abu, Brahim Bensaou, **Ahmed M. Abdelmoniem**, **“Dimensioning the pending interest table in content-centric networks”**. *Future Generation Computer Systems (FGCS), Elsevier, Volume 152, 2024, Pages 179-192*
- Nauman Khan, Rosli bin Salleh, Zahid Khan, Anis Koubaa, Mosab Hamdan, **Ahmed M. Abdelmoniem**, **“Ensuring Reliable Network Operations and Maintenance: The Role of PMRF for Switch Maintenance and Upgrades in SDN”**. *Journal of King Saud University - Computer and Information Sciences, Elsevier, 2023*
- Karan Gadgil, Sukhpal Singh Gill, Ahmed M. Abdelmoniem, **“A Meta-learning based Stacked Regression Approach for Customer Lifetime Value Prediction”**. *Journal of Economy and Technology, Elsevier, 2023*
- Eric Bragion, Habiba Akter, Mohit Kumar, Minxian Xu, **Ahmed M. Abdelmoniem**, Sukhpal Singh Gill, **Fortaleza: The emergence of a network hub**. *Internet of Things and Cyber-Physical Systems, Volume 3, 2023, Pages 272-279*
- Sundas Iftikhar, Sukhpal Singh Gill, Chenghao Song, Minxian Xu, Mohammad Sadegh Aslanpour, Adel N Toosi, Junhui Du, Huaming Wu, Shreya Ghosh, Deepraj Chowdhury, Muhammed Golec, Mohit Kumar, **Ahmed M Abdelmoniem**, Felix Cuadrado, Blesson Varghese, Omer Rana, Schahram Dustdar, Steve Uhlig. **“AI-based fog and edge computing: A systematic review, taxonomy and future directions”**. *Internet Of Things Journal, Elsevier, 2022.*
- S Abdulah, W Atwa, **Ahmed M. Abdelmoniem**. **Active clustering data streams with affinity propagation**. *ICT Express 8 (2), 276-282, 2022*
- **Ahmed M. Abdelmoniem** and Brahim Bensaou. **“T-RACKs: A Faster Recovery Mechanism for TCP in Data Center Networks”**. *ACM/IEEE Transactions on Networking (ToN), 2021.*
- Jiaqing Dong, Chen Tian, **Ahmed M. Abdelmoniem**, Huaping Zhou, Bo Bai, Hao Yin, Gong Zhang. **“Uranus: Congestion-proportionality among Slices based on Weighted Virtual Congestion Control”**. *Computer Networks, Elsevier.*
- **Ahmed M. Abdelmoniem**, Brahim Bensaou, and Amuda James Abu. **“Mitigating Incast-TCP Congestion in Data Centers with SDN”**. *Annals of Telecommunications - Springer. Special issue on Cloud Communications and Networking.*
- **Ahmed M. Abdelmoniem**, Hosny M. Ibrahim, Marghny H. Mohamed, and Abdel-Rahman Hedar. **“Ant Colony and Load Balancing Optimizations for AODV Routing Protocol”**. *International Journal of Sensor Networks and Data Communications*, volume 1, 2011. doi:10.4303/ijsndc/X110203

Published International Refereed Conference Publications

- **Ahmed M. Abdelmoniem**, AN Sahu, M Canini, SA Fahmy. **“REFL: Resource Efficient Federated Learning”**. *In Proceedings of ACM EuroSys, Rome, Italy, 2023*
- **Ahmed M. Abdelmoniem**, Y. M. Abdelmoniem and A. Elzanaty, **“A2FL: Availability-Aware Selection for Machine Learning on Clients with Federated Big Data”**, *In Proceedings of IEEE ICC, Rome, Italy, 2023*
- Chehri Abdellah, **Ahmed M. Abdelmoniem**, Muhammad Zeeshan Shakir, **“Machine Learning Classification of Hermite Gaussian Beams for 5G and Beyond Free-Space Optical Backhaul Links”**. *In Proceedings of IEEE Globecom, Kuala Lumpur, Malaysia, 2023*
- Tianfan Zhang, Huaping Zhou, Chengyuan Huang, Chen Tian, Wei Zhang, Xiaoliang Wang, Yi Wang, **Ahmed M. Abdelmoniem**, Matthew Tan, Wanchun Dou, Guihai Chen. **“Achieving Zero-copy Serialization for Datacenter RPC”**. *In Proceedings of IEEE IPCCC, Anaheim, CA, USA, 2023*
- Moudy Alshareef, Mona Jaber, **Ahmed M. Abdelmoniem**. **“A Differential Privacy Approach for Privacy-Preserving Multi-Modal Stress Detection”**. *In Proceedings of IEEE CAMAD, Edinburgh, UK, 2023*
- Sofia Zahri, Hajar Bennouri, Abdellah Chehri, **Ahmed M. Abdelmoniem**. **“Federated Learning for IoT Networks: Enhancing Efficiency and Privacy”**. *In Proceedings of IEEE World Forum on IoT, Portugal, 2023*
- Amna Arouj, **Ahmed M. Abdelmoniem**. **Towards Energy-Aware Federated Learning on Battery-Powered Clients**. *In Proceedings of the ACM Workshop on Data Privacy and Federated Learning Technologies for Mobile Edge Networks (FedEdge), ACM MobiCom, 2022*
- **Ahmed M. Abdelmoniem**. **Towards Efficient and Practical Federated Learning**. *Proceedings of the Workshop on Cross-Community Federated Learning: Algorithms, Systems and Co-designs (CrossFL), MLSys, 2022*
- **Ahmed M. Abdelmoniem**, CY Ho, P Papageorgiou, M Canini. **Empirical analysis of federated learning in heterogeneous environments**. *Proceedings of the 2nd European Workshop on Machine Learning and Systems (EuroMLSys), ACM EuroSys, 2022*
- Atal Sahu, Aritra Dutta, **Ahmed M Abdelmoniem**, Trambak Banerjee, Marco Canini, Panos Kalnis. **Rethinking gradient sparsification as total error minimization**. *Proceedings of NeurIPS, Spotlight (Top 3%), Virtual Conference, 2022.*
- Kelvin H.T. Chiu, Jason Min Wang, **Ahmed M. Abdelmoniem**, Brahim Bensaou. **“A Two-tiered Caching Scheme for Information-Centric Networks”**. *Proceedings of IEEE High Performance Switching and Routing (IEEE HPSR), Paris, France, June 2021.*
- **Ahmed M. Abdelmoniem**, Marco Canini. **“DC2: Delay-aware Compression Control for Distributed Machine Learning”**. *Proceedings of IEEE Computer Communications Conference (IEEE INFOCOM), Virtual Conference, May 2021.*
- **Ahmed M. Abdelmoniem**, Marco Canini. **“Towards Mitigating Device Heterogeneity in Federated Learning via Adaptive Model Quantization”**. *Proceedings of EuroMLSys workshop at ACM European Conference on Computer Systems (ACM EuroSys), Virtual Conference, Apr 2021.*
- Hang Xu, Chen yu-ho, **Ahmed M. Abdelmoniem**, Aritra Dutta, Elhoucine Bergou, Konstantinos Karatsenidis, Marco Canini, Panos Kalnis, **“GRACE: A Compressed Communication Framework for Distributed Machine Learning”**. *Proceedings of IEEE International Conference on Distributed Computing Systems (IEEE ICDCS), Virtual Conference, 2021.*
- **Ahmed M. Abdelmoniem**, Ahmed Elzanaty, Mohamed Slim-alouini, Marco Canini. **“An Efficient Statistical-based Gradient Compression Technique for Distributed Training Systems”**. *Proceedings of the International Conference on Machine Learning and Systems (MLSys), Virtual Conference, Apr 2021.*

- Rishikesh R. Gajjala*, Shashwat Banchhor*, **Ahmed M. Abdelmoniem***, Aritra Dutta, Marco Canini, Panos Kalnis. **"Huffman Coding Based Encoding Techniques for Fast Distributed Deep Learning"**. *Proceedings of Distributed ML workshop at 16th ACM International Conference on emerging Networking EXperiments and Technologies (ACM CoNEXT), Virtual Conference, Dec 2020.*
- **Ahmed M. Abdelmoniem**, Brahim Bensaou, Hengky Susanto. **"Reducing Latency in Multi-Tenant Data Centers via Cautious Congestion Watch"**. *Proceedings of 49th ACM International Conference on Parallel Processing - ICPP, Edmonton, Canada, 2020.*
- Arritra Dutta, Houcine Bergou, **Ahmed M. Abdelmoniem**, Chen-yu Ho, Atal Sahu, Marco Canini, Panos Kalnis, **"On the Discrepancy between the Theoretical Analysis and Practical Implementations of Compressed Communication for Distributed Deep Learning"**. *Proceedings of Thirty-Forth AAAI Conference on Artificial Intelligence (AAAI-20), New York, USA, Feb 2020.*
- **Ahmed M. Abdelmoniem**, Brahim Bensaou and Hengky Susanto. **"Taming Latencies in Data Center Networks via Active Congestion-Probing"**. *Proceedings of IEEE International Conference on Distributed Computing Systems (ICDCS), Dallas, Texas, USA, July 2019.*
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- **Ahmed M. Abdelmoniem"DC2: Delay-aware Compression Control for Distributed Machine Learning"**. *Proceedings of IEEE Computer Communications Conference (IEEE INFOCOM), Virtual Conference, May 2021.*
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- **Ahmed M. Abdelmoniem**. **"Towards Practical and Efficient Federated Learning"**. *The Intelligent Methods, Systems, and Applications (IMSA) Conference, Cairo, Egypt, Jul. 2023*
- **Ahmed M. Abdelmoniem**. **"AI and Edge Technologies for Fostering SDGs"**. *International Conference on SUSTAINABILITY: Recent developments in research and teaching towards the United Nations Sustainable Development Goals, Online, Jun. 2023*
- **Ahmed M. Abdelmoniem**. **"REFL: Resource-Efficient Federated Learning"**. *MobiUK workshop, Lancaster, UK, May. 2023*
- **Ahmed M. Abdelmoniem**. **"Distributed Deep Learning Clinic"**. *KAUST-NeurIPS meet-up workshop, Saudi Arabia, Dec. 2019*
- **Ahmed M. Abdelmoniem**. **"Data Analytics in the Cloud (hands-on tutorial)"**. *The BioDialog Project: Exhibition and Hackathon on BioDiversity Informatics, Assiut University, Egypt, Nov. 2018*
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